Enzymes & Temperature

Exam code: 0610 & 0970

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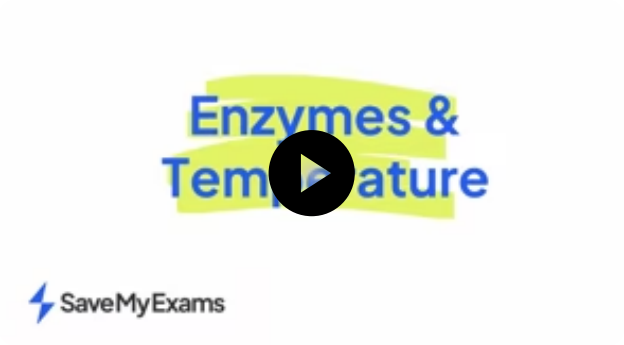

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Help me understand

Enzymes & Temperature: Extended

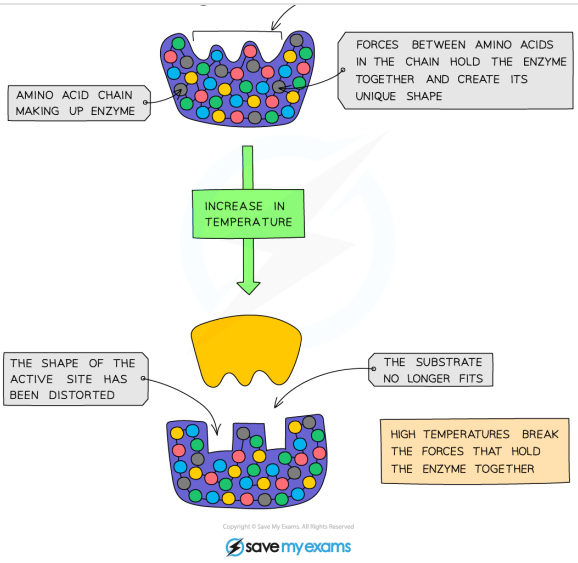
Extended Tier Only

- Enzymes are **proteins** and have a **specific shape**, held in place by **bonds**
- This is extremely important around the **active site** area as the specific shape is what ensures the **substrate will fit into the active site** and enable the reaction to proceed
- Enzymes work fastest at their '**optimum temperature**' – in the human body, the optimum temperature is 37ºC
- Heating to high temperatures (beyond the optimum) will **break the bonds** that hold the enzyme together and it will lose its shape -this is known as **denaturation**
- Substrates cannot fit into denatured enzymes as the shape of their active site has been lost
- Denaturation is **irreversible** - once enzymes are denatured they cannot regain their proper shape and activity will stop

Stuck on something?

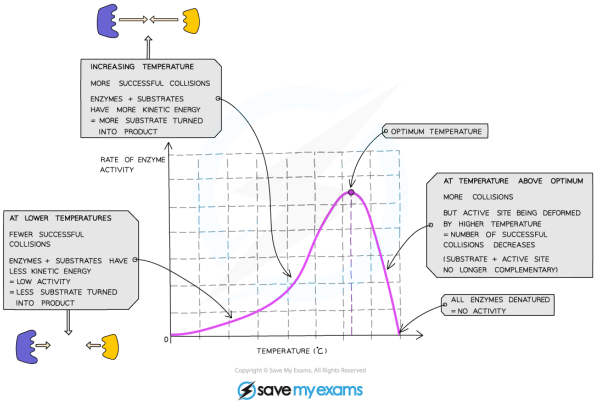
Ask questions and get help with exactly what you're revising.

Got it



Effect of temperature on enzyme activity

- Increasing the temperature from 0ºC to the optimum increases the activity of enzymes as **the more energy the molecules have the faster they move and the number of collisions with the substrate molecules increases**, leading to a faster rate of reaction
- This means that **low temperatures do not denature enzymes**, they just make them work more slowly



Graph showing the effect of temperature on the rate of enzyme activity

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